# Results: Language, Topics, and Embeddings

## RQ3: Topical and Linguistic Structure I

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 500-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: modafinil, treatment, study, results, patients, effects, sleep, used, clinical, use, drug, studies, mg, using, sleepiness, disorder, associated, cognitive, narcolepsy, significant.

**Table 3. NMF topics extracted from the corpus.**

# RQ3: Topical and Linguistic Structure I

| Topic | Top terms |
|---|---|
| 0 | modafinil, mg, kg, effects, rats, mice, induced, sleep |
| 1 | fatigue, patients, placebo, modafinil, scale, armodafinil, treatment, depression |
| 2 | h4, methods, results, conclusion, 95, analysis, ci, risk |
| 3 | use, cognitive, drugs, adhd, studies, drug, cocaine, methylphenidate |
| 4 | sleep, narcolepsy, sleepiness, cataplexy, daytime, patients, eds, excessive |

## RQ3: Topical and Linguistic Structure I

The topics reveal the thematic structure of the literature: Topic 0 reflects preclinical dose-response pharmacology in animal models (mg/kg dosing, dopamine activity); Topic 1 captures clinical fatigue and mood trials (placebo-controlled patient studies, armodafinil); Topic 2 collects structured-abstract and meta-analytic language (methods, results, 95% confidence intervals, risk estimates); Topic 3 centres on cognitive enhancement and psychostimulant pharmacology (ADHD, methylphenidate, cocaine comparisons); and Topic 4 addresses sleep disorders (narcolepsy, excessive daytime sleepiness, cataplexy, hypersomnia). These topics cross-cut the keyword-based subfield taxonomy, revealing connections that the explicit classification does not capture.

# RQ3: Topical and Linguistic Structure I

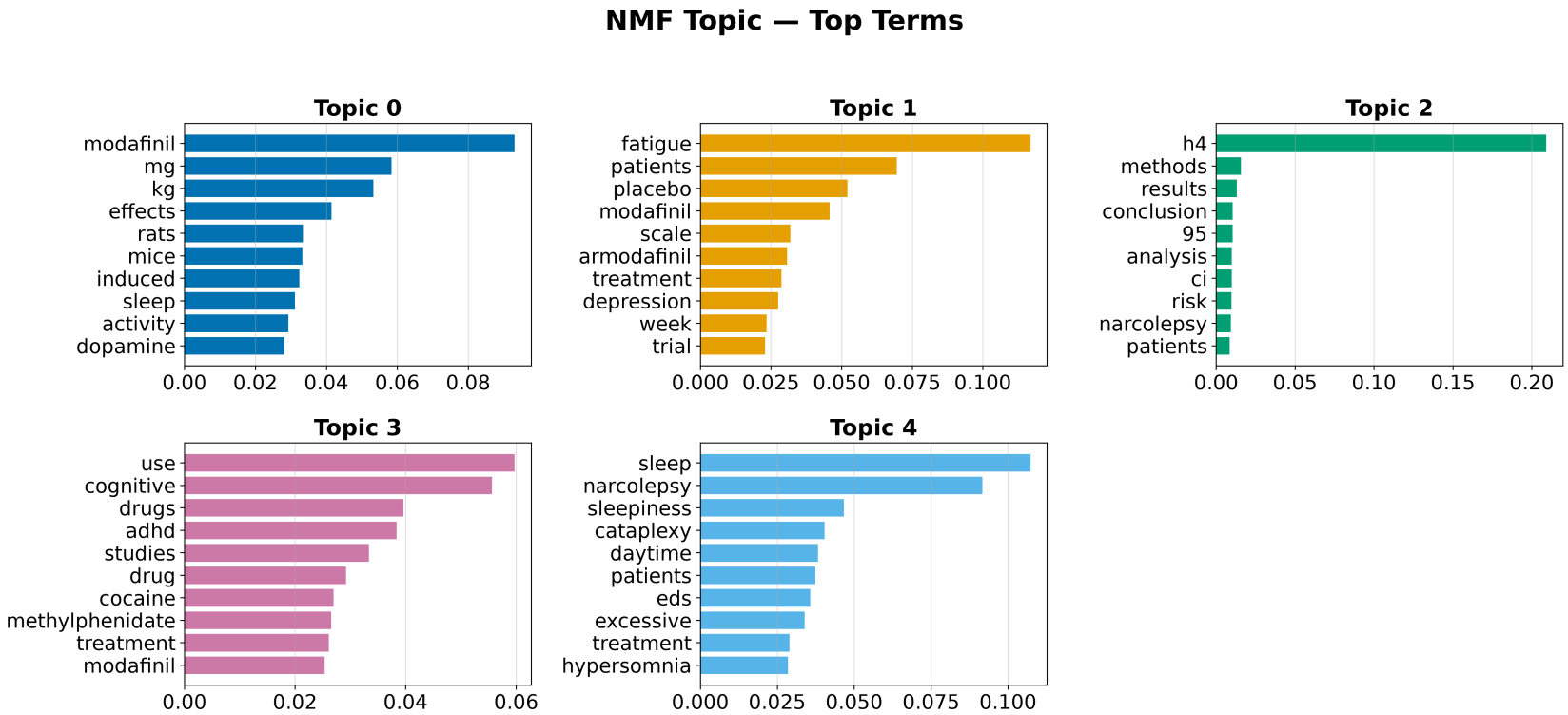


Figure 9: Topic-term bar charts for Modafinil. Each panel shows the top weighted terms for one of 5 NMF topics, with bar length proportional to the topic-term weight in the $\mathbf{H}$ matrix.

## Document Embeddings I

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

# Document Embeddings I

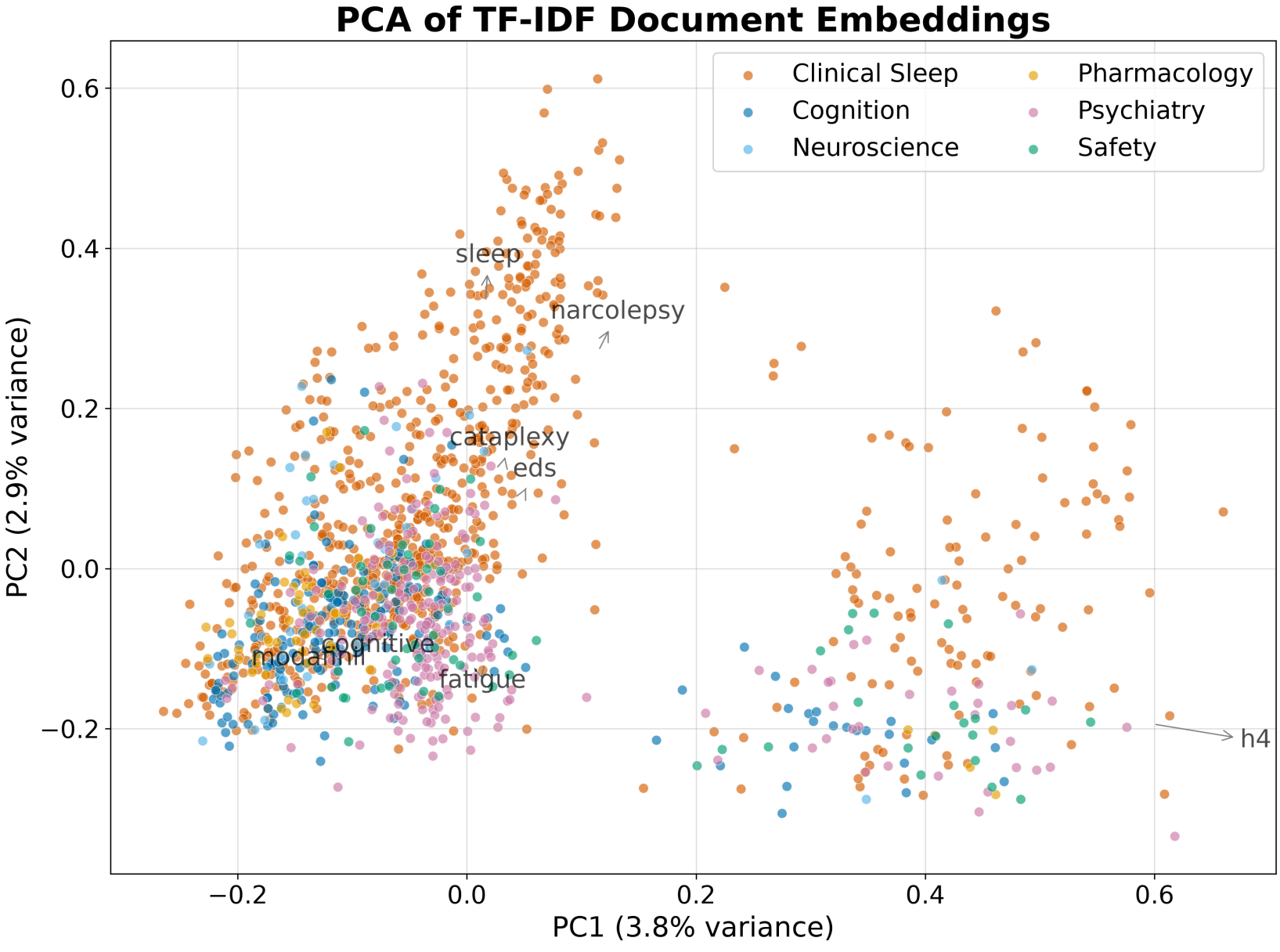


Figure 10: PCA projection of document embeddings for Modafinil. Each point represents one document projected onto the first two principal components of the TF-IDF/SVD embedding. Colours indicate subfield assignment, showing how the topical geography relates to the keyword taxonomy.

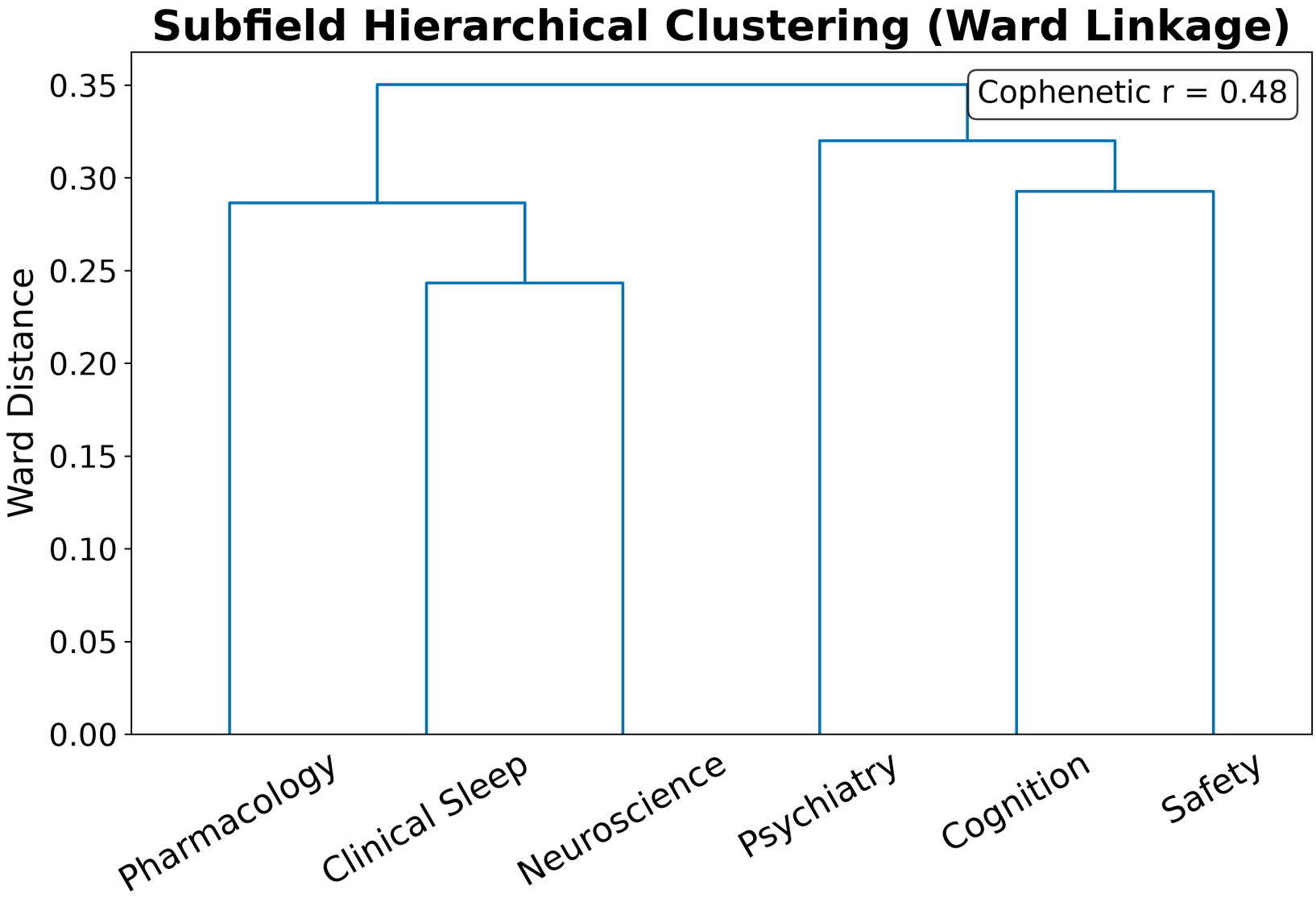

Subfield Hierarchical Clustering (Ward Linkage)
Cophenetic r = 0.48
Ward Distance
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Pharmacology
Clinical Sleep
Neuroscience
Psychiatry
Cognition
Safety

Figure 11: Hierarchical clustering dendrogram of subfield TF-IDF centroids. Leaves are the configured subfields; the tree shows the similarity structure between subfield centroid vectors.

# Term Analysis I

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.

# Term Analysis I

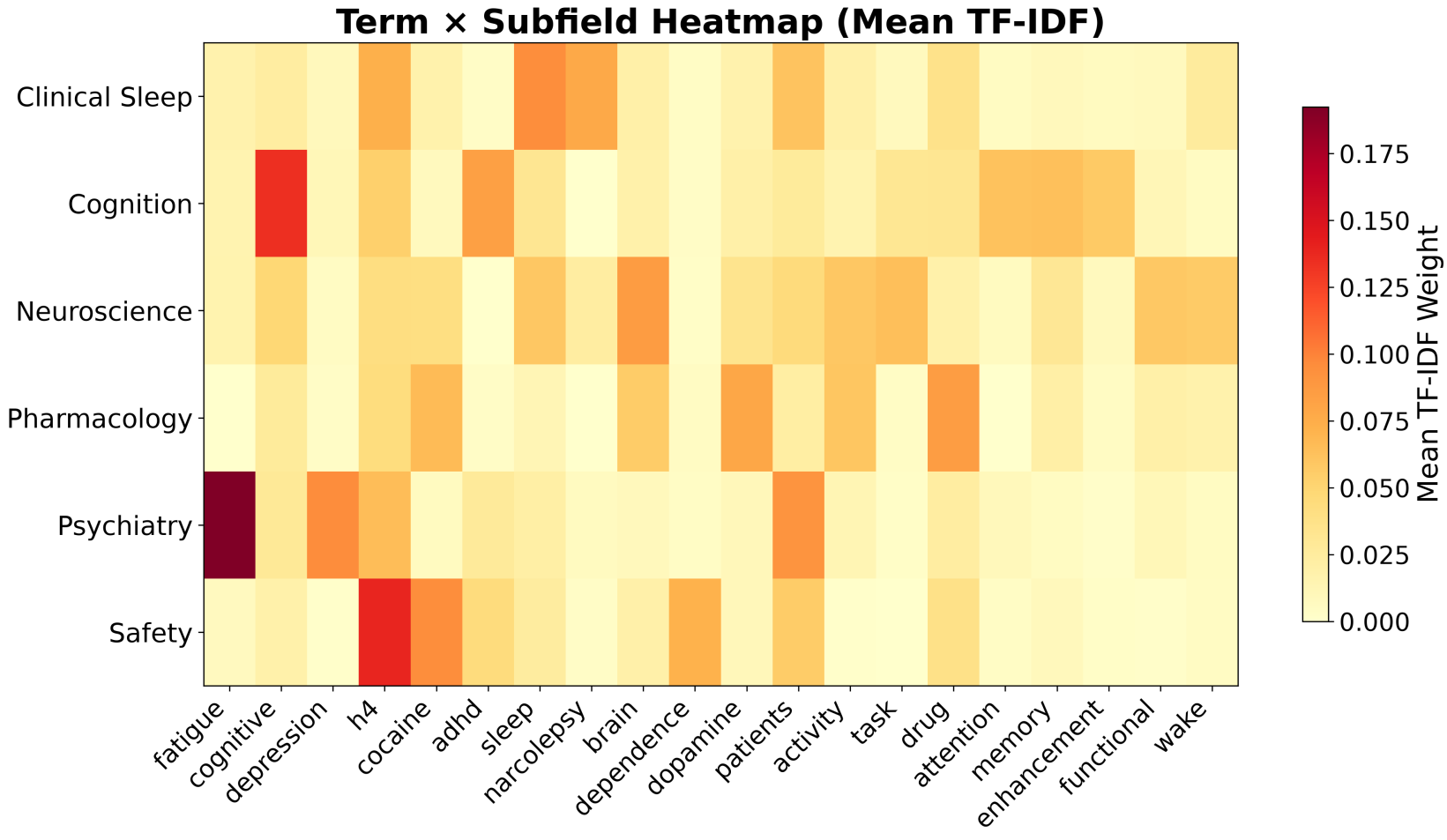


Figure 12: Term heatmap for Modafinil. Each cell shows the mean TF-IDF weight of a term within a subfield. Terms are selected by between-subfield variance to highlight discriminative vocabulary rather than globally frequent terms.

Named entity extraction over the 1437 abstracts identified 30 unique entities. The most frequent entities reflect the clinical and pharmacological vocabulary of the modafinil literature.

**Table 4. Top named entities in abstracts.**

# Named Entity Analysis I

| Entity | Frequency |
|---|---|
| ADHD | 392 |
| CI | 343 |
| EDS | 326 |
| OSA | 272 |
| MOD | 181 |
| ESS | 148 |
| RESULTS | 148 |
| MS | 145 |

# Named Entity Analysis I

| Entity | Frequency |
|---|---|
| DAT | 134 |
| NT1 | 131 |
| SD | 130 |
| CONCLUSIONS | 129 |
| MD | 102 |
| CE | 101 |
| IH | 101 |

# Named Entity Analysis I

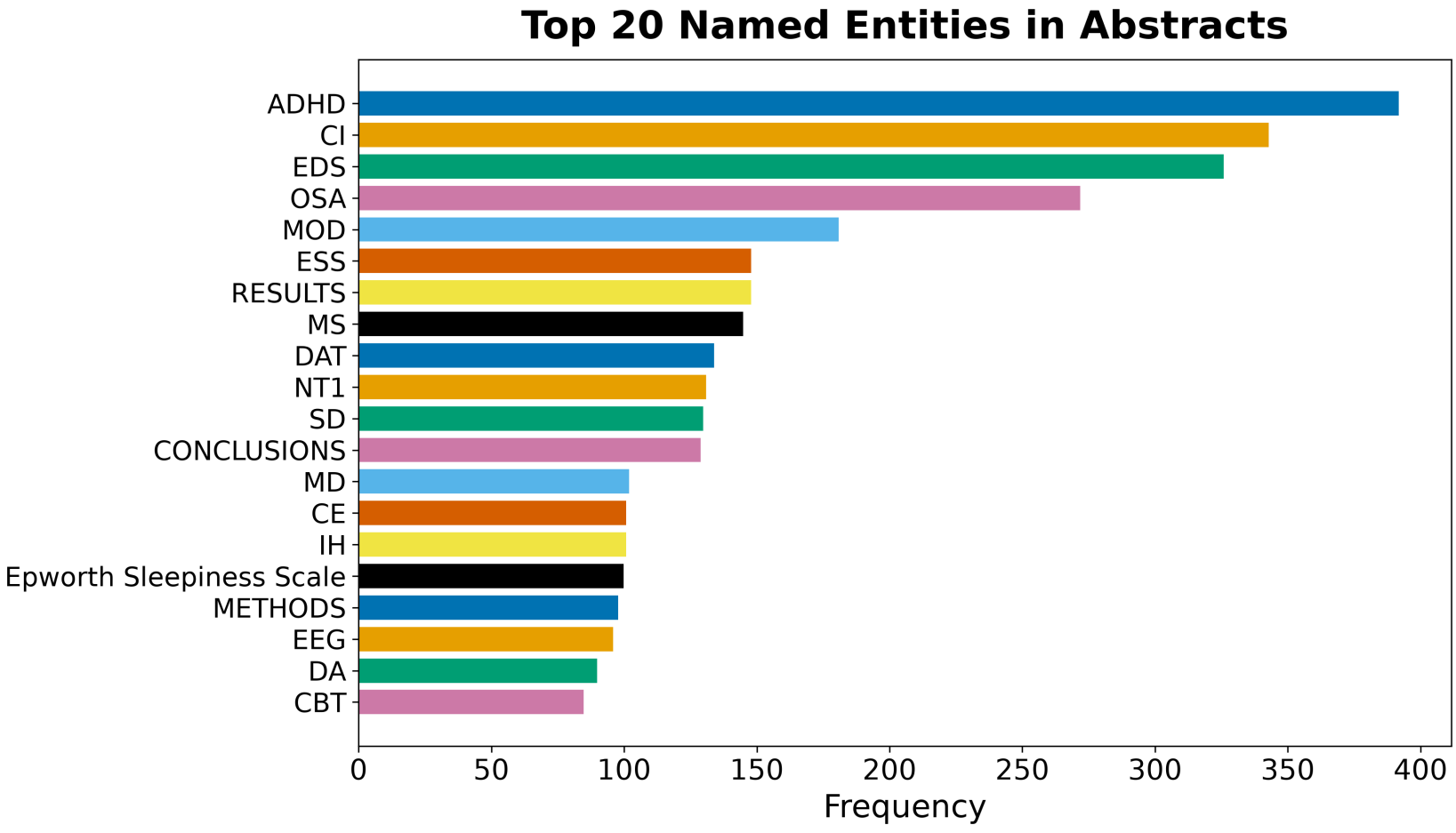


Figure 13: Top named entities for Modafinil. The horizontal bar chart shows the 20 most frequently extracted named entities from abstracts, revealing the dominant drugs, conditions, and concepts in the literature.

**Table 5. Top keyphrases by TF-IDF score.**

# Named Entity Analysis I

| Keyphrase | Score |
|---|---|
| available | 0.3333 |
| abstract | 0.3333 |
| abstract available | 0.3333 |
| content | 0.1053 |
| access | 0.1053 |
| sub | 0.0881 |
| md | 0.0870 |
| jama | 0.0826 |

# Named Entity Analysis I

| Keyphrase | Score |
|---|---|
| cleveland | 0.0763 |
| modafinil | 0.0741 |
| depression | 0.0741 |
| cognitive | 0.0714 |
| substance | 0.0694 |
| drug | 0.0667 |
| conditions | 0.0667 |

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

**Table 6. Top 10 most similar document pairs.**

# Embedding Similarity and Clustering I

| Paper A | Paper B | Similarity |
|---|---|---|
| doi:10.1176/appi.ajp.163.12.2184 | doi:10.1176/ajp.2006.163.12.2184 | 1.0000 |
| doi:10.1176/ajp.2006.163.12.2184 | doi:10.1176/appi.ajp.163.12.2184 | 1.0000 |
| doi:10.1345/aph.1h302 | doi:10.1136/bcr.08.2011.4652 | 0.9660 |
| doi:10.4088/jcp.09m05900gry | doi:10.1186/s40345-015-0034-0 | 0.9566 |
| doi:10.1111/bdi.12859 | doi:10.1111/acps.12712 | 0.9558 |
| doi:10.1111/j.1360-0443.2008.02437.x | doi:10.1111/j.1465-3362.2012.00473.x | 0.9536 |
| doi:10.1513/annalsats.202006-696fr | doi:10.3760/cma.j.cn112147-20241118-00681 | 0.9532 |
| doi:10.1164/ajrccm.164.9.2103032 | doi:10.1093/sleep/28.4.464 | 0.9513 |

# Embedding Similarity and Clustering I

| Paper A | Paper B | Similarity |
|---|---|---|
| doi:10.1093/sleep/28.4.464 | doi:10.1164/ajrccm.164.9.2103032 | 0.9513 |
| doi:10.1002/cncr.25083 | doi:10.1200/jco.2005.23.16_suppl.8012 | 0.9507 |

# Embedding Similarity and Clustering I

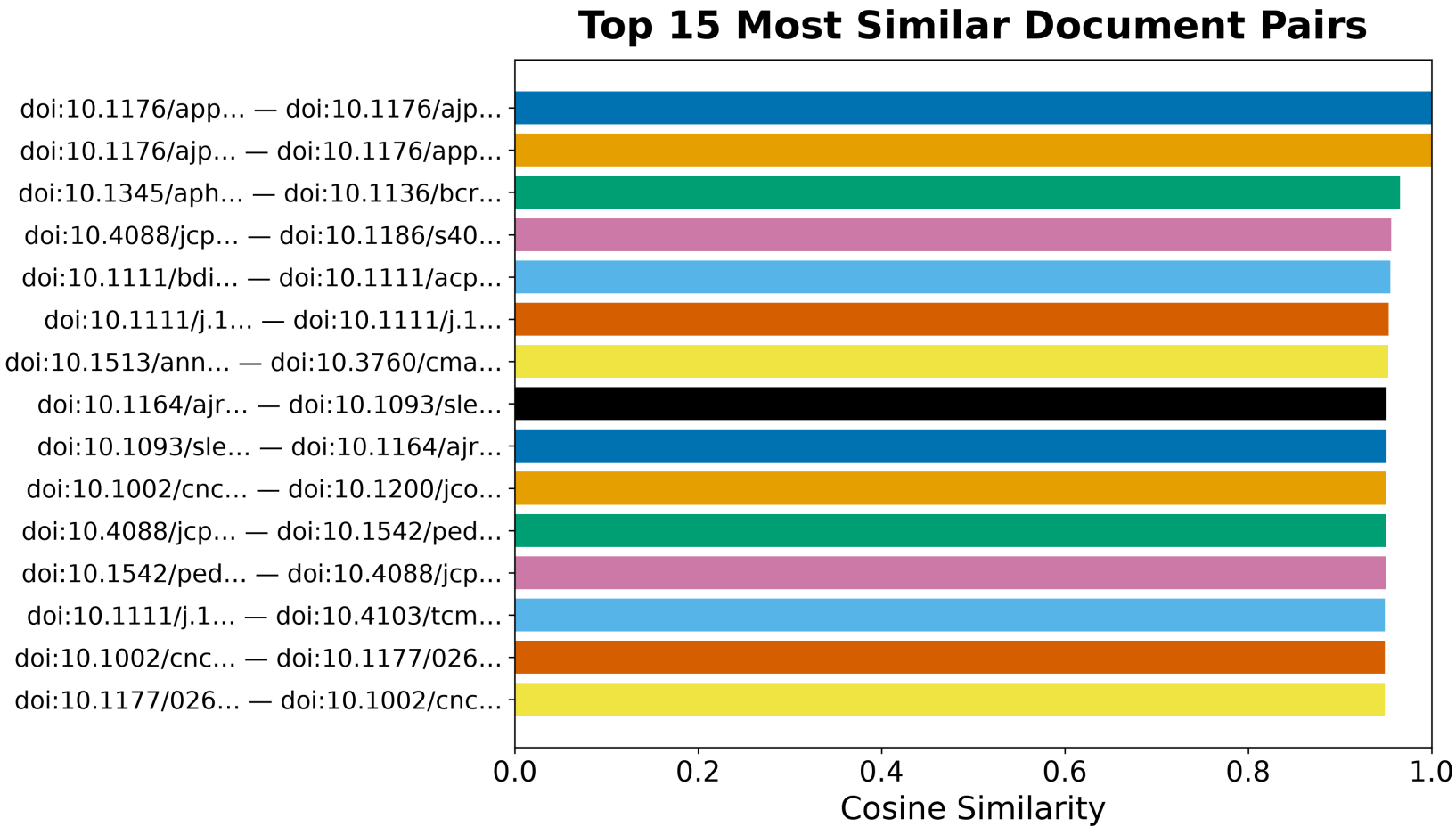


Figure 14: Document similarity for Modafinil. The horizontal bar chart shows the 15 most similar document pairs ranked by cosine similarity of their TF-IDF/SVD embeddings. High-similarity pairs share topical and lexical content.

# Embedding Similarity and Clustering I

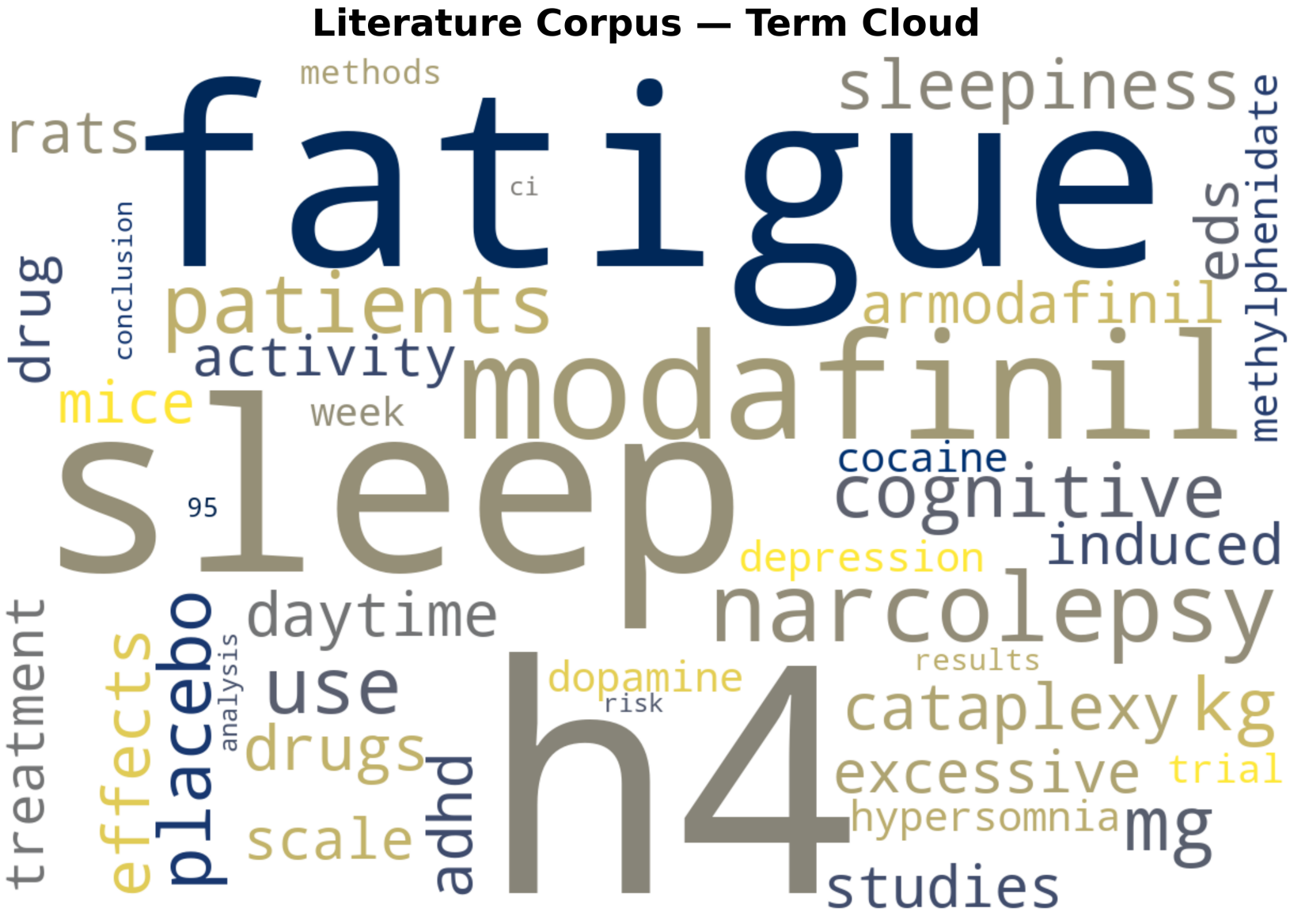


Figure 15: Term cloud for Modafinil. Term sizes are proportional to their topic-term weights from the discovered topics, providing a visual summary of the dominant vocabulary.

# Embedding Similarity and Clustering I

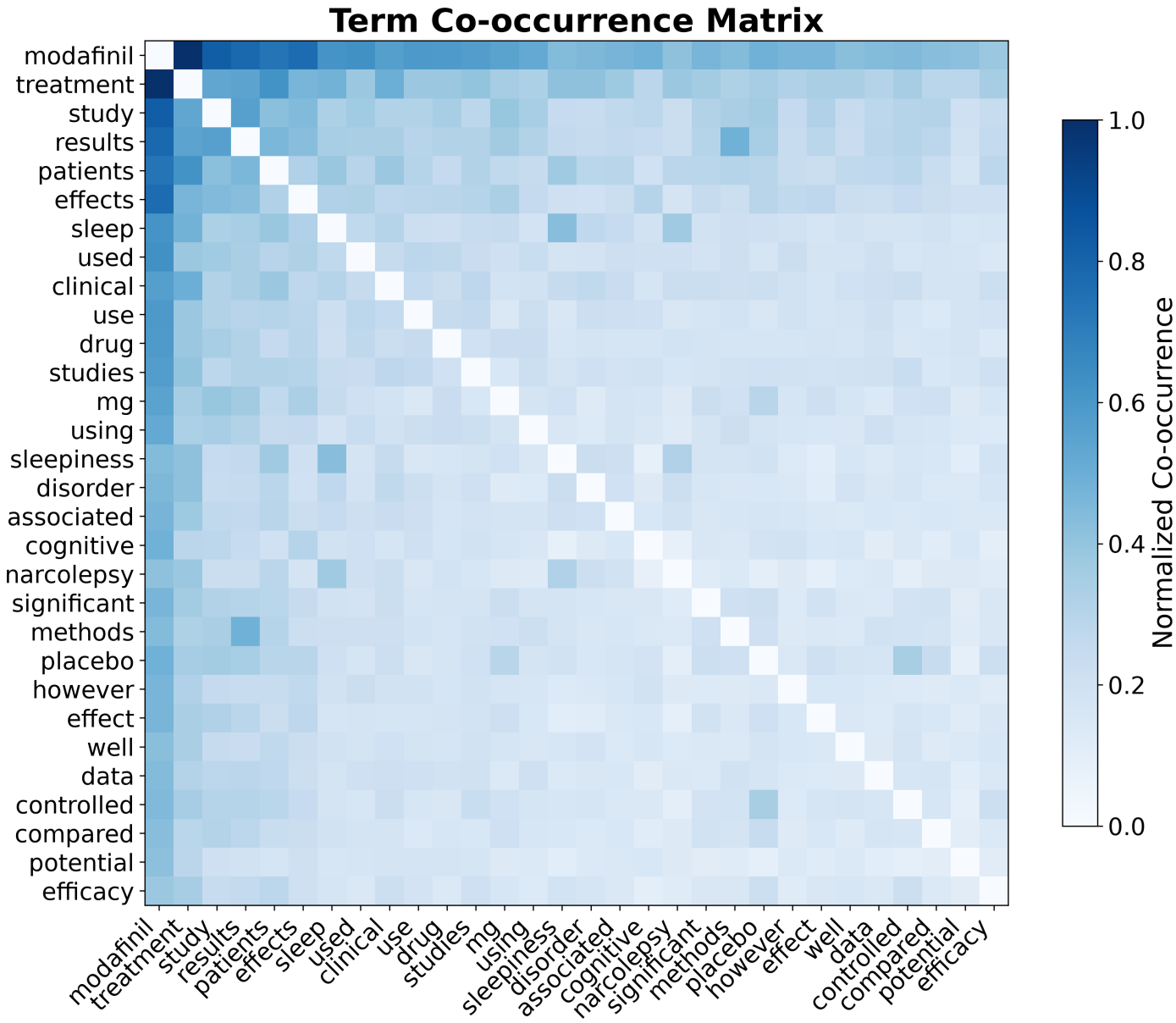


Figure 16: Term co-occurrence matrix for Modafinil. Each cell shows the normalized co-occurrence frequency of two terms within the same document, revealing which concepts tend to appear together in the literature.

## Embedding Similarity and Clustering I

These embeddings support semantic retrieval over the corpus and the visual map of the literature's topical geography.